

Problem Set 11

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Variance

Problem 1: Variance of a Bernoulli trial.

Let X be a Bernoulli random variable with parameter $p \in [0, 1]$:

$$X = \begin{cases} 1 & \text{with probability } p, \\ 0 & \text{with probability } 1 - p. \end{cases}$$

- (a) Compute $\mathbb{E}[X]$ and $\mathbb{E}[X^2]$. *Hint: since $X \in \{0, 1\}$, note that $X^2 = X$.*
- (b) Using $\text{Var}[X] = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$, show that

$$\text{Var}[X] = p(1 - p).$$

- (c) Show that $\text{Var}[X] \leq \frac{1}{4}$ for every $p \in [0, 1]$, and identify the value of p that attains this maximum. *Hint: maximize $f(p) = p(1 - p)$ over $[0, 1]$.*

Problem 2: Variance of the Binomial distribution.

Let $Y \sim \text{Binomial}(n, p)$. Equivalently, write $Y = \sum_{i=1}^n X_i$, where $X_1, \dots, X_n$ are **independent** Bernoulli(p) random variables (as in Problem 1).

- (a) Using linearity of expectation, show that $\mathbb{E}[Y] = np$.
- (b) Using independence and your answer to Problem 1(b), show that

$$\text{Var}[Y] = np(1 - p).$$

State clearly where you use independence.

- (c) Conclude that $\text{Var}[Y] \leq \frac{n}{4}$ for every $p \in [0, 1]$.

MinHash

Problem 3: How many hash functions does MinHash need?

Recall the MinHash estimator for the Jaccard similarity $J = J(A, B)$ of two sets. We draw k independent hash functions $h_1, \dots, h_k$ and set

$$X_i = \begin{cases} 1 & \text{if } \min_{x \in A} h_i(x) = \min_{x \in B} h_i(x), \\ 0 & \text{otherwise,} \end{cases} \quad \hat{J} = \frac{1}{k} \sum_{i=1}^k X_i.$$

Each X_i is an independent Bernoulli(J) random variable, i.e. $\Pr[X_i = 1] = J$.

- (a) Identify the distribution of $\sum_{i=1}^k X_i = k\hat{J}$ (the number of agreeing hash functions). Using Problems 1 and 2, state $\mathbb{E}[\hat{J}]$ and $\text{Var}[\hat{J}]$ in terms of J and k .
- (b) Apply **Chebyshev's inequality** to bound

$$\Pr[|\hat{J} - J| \geq 0.05]$$

in terms of k . Use the bound $\text{Var}[X_i] \leq \frac{1}{4}$ from Problem 1(c) so that your answer does not depend on the unknown J .

- (c) How many hash functions k are needed to guarantee an additive error of at most ± 0.05 (that is, $|\hat{J} - J| \leq 0.05$) with probability at least 0.95? Report a concrete number.